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## Lessons



# The Distributive Property

The **distributive property** involves multiplication and addition. Study this example to understand how the property works.

### Example 1 Hide

At a dollar store, all items are $1.10. Maia buys 7 plastic tablecloths and 3 packs of paper plates. How much money did she spend?

| Method 1 | Method 2 |
|---|---|
| Total items: $7 + 3 = 10$ | Cost of tablecloths: $7 \times \$1.10 = \$7.70$ |
| Multiply by cost: $10 \times \$1.10 = \$11.00$ | Cost of plates: $3 \times \$1.10 = \$3.30$ |
| | Total cost: $\$7.70 + \$3.30 = \$11.00$ |

In the example, both methods gave a final answer of $11. How would you describe the difference between the methods? Method 1 adds first. Method 2 multiplies first. The distributive property describes this relationship between adding and multiplying.

Instead of adding $b$ and $c$ and then multiplying by $a$, you can distribute $a$ to both $b$ and $c$ and then add.

$$a(b + c) = ab + ac$$

***Note:*** Writing two variables next to each other indicates multiplication: $a(b + c)$ means $a \times (b + c)$, and $ab + ac$ means $(a \times b) + (a \times c)$.

You can also use the distributive property with subtraction.

$$a(b - c) = ab - ac$$

### Example 2 Hide

Carol used to earn $11 per hour. Now she earns $16 per hour. How much more does she earn now for a 40-hour work week?

| Method 1 | Method 2 |
|---|---|
| $40 \times (\$16 - \$11)$ | $(40 \times \$16) - (40 \times \$11)$ |
| $= 40 \times \$5$ | $= \$640 - \$440$ |
| $= \$200$ | $= \$200$ |

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